

PROJECT REPORT

Journal-entry testing, decided.

JE Agent — Journal Entry Testing AI Agent
per ISA 240 / AS 2401

PREPARED BY

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Journal Entry Testing, Decided.

JE Agent, an AI-assisted audit assistant for journal entries (ISA 240 / AS 2401)

Every audit season the same scene plays out somewhere. An auditor opens a client's general ledger export, and two hundred thousand journal entries pour across the screen. Debits and credits, dates and account numbers, an endless gray river of bookkeeping. And yet somewhere in that river swim a few dozen entries that were never quite honest: an invoice sliced into pieces to slip under an approval limit, a round number invented behind a desk, a reversal posted the day after a bonus was paid.

This report tells the story of JE Agent, a tool built to find those entries. It explains the idea, the way the system thinks, and, most importantly, what happened when we tested whether its promises survive contact with measured evidence. No technical background is needed to follow along. The deep detail lives in the repository; this document is the story and the proof.

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1 · The Problem

Standards such as ISA 240 and AS 2401 require auditors to test journal entries for fraud risk. The requirement is easy to write and hard to live. Anyone who has faced a full-year export knows the three ways this job actually gets done, and why each one disappoints.

The first way is **by hand**. A senior opens the ledger and reads. Experience helps, attention fails. Two hundred thousand lines cannot be read by a human being in any honest sense of the word “read”, so sampling takes over, and whatever the sample misses is missed silently.

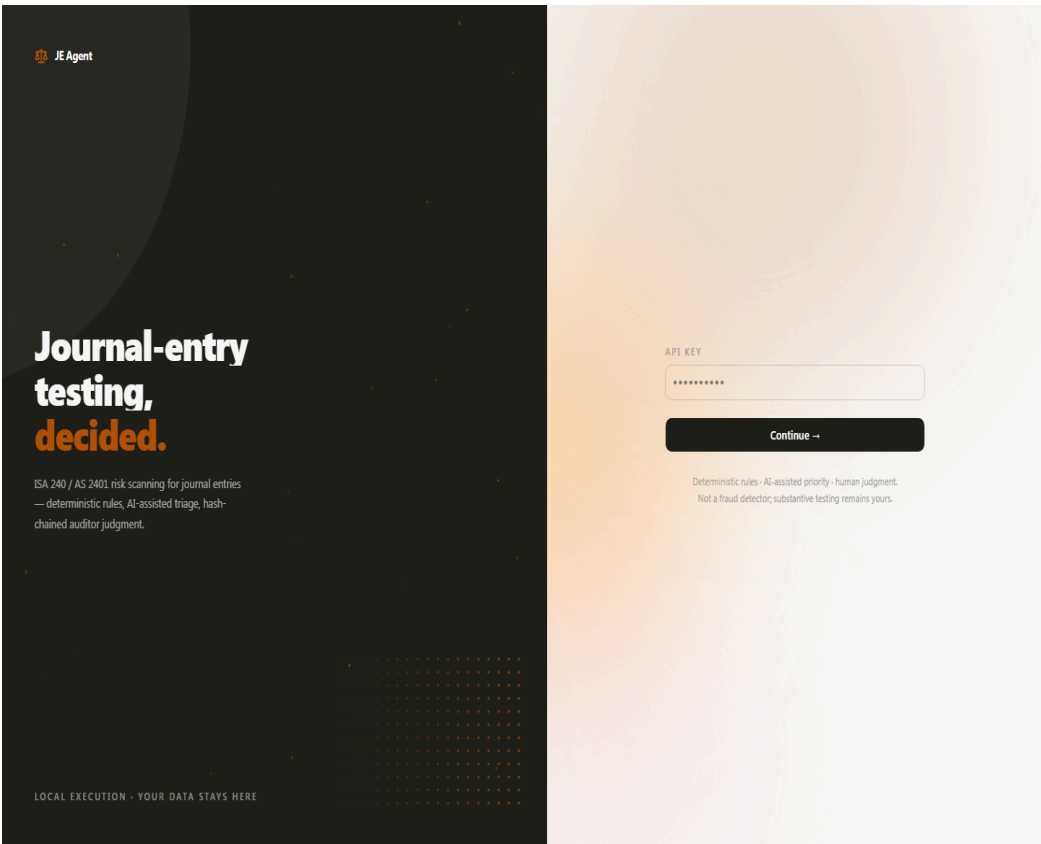
The second way is **with a script**. Someone writes a filter: show me every round amount above fifty thousand. The script returns four thousand hits, and almost all of them are innocent. Big real invoices happen to be round. Month-end posting is normal business, not a crime. The auditor drowns in noise and stops trusting the tool.

The third way is **with AI alone**. Hand the ledger to a model and ask it to find fraud. It will confidently point at entries, some of them genuinely odd, others pure invention, and none of them carrying accountability. An auditor who cannot trace *why* an entry was flagged cannot defend the workpaper. A black box does not sign off.

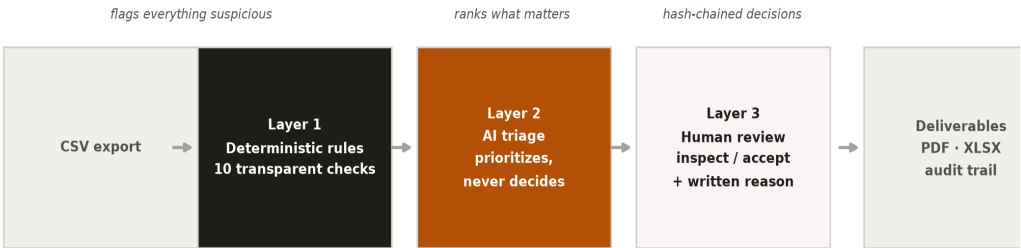
Each approach fails differently: humans miss volume, scripts drown in false alarms, models cannot take responsibility. The insight behind JE Agent is that this is not a reason to despair. It is a division of labor waiting to happen. Let the machine do

what machines do best, let the model do what models do best, and put the decision where it always belonged, with a person who signs their name.

2 · The Logic, Three Layers



JE Agent receives an accounting export and passes it through three layers, each one trusted exactly as far as it deserves.



Layer 1, deterministic rules. Ten transparent checks sweep every single entry: round amounts, splitting, period-end timing, balance failures, rare users, odd account

pairings, reversals, date divergence, high-risk users, manual-entry concentration. Nothing here is intelligent, and that is the point. A rule is a sentence anyone can read, argue with, and audit. When it fires, it says precisely why. These rules never get tired and never miss the pattern they encode. But they also cry wolf, as we will see in Section 4. They cast a wide net.

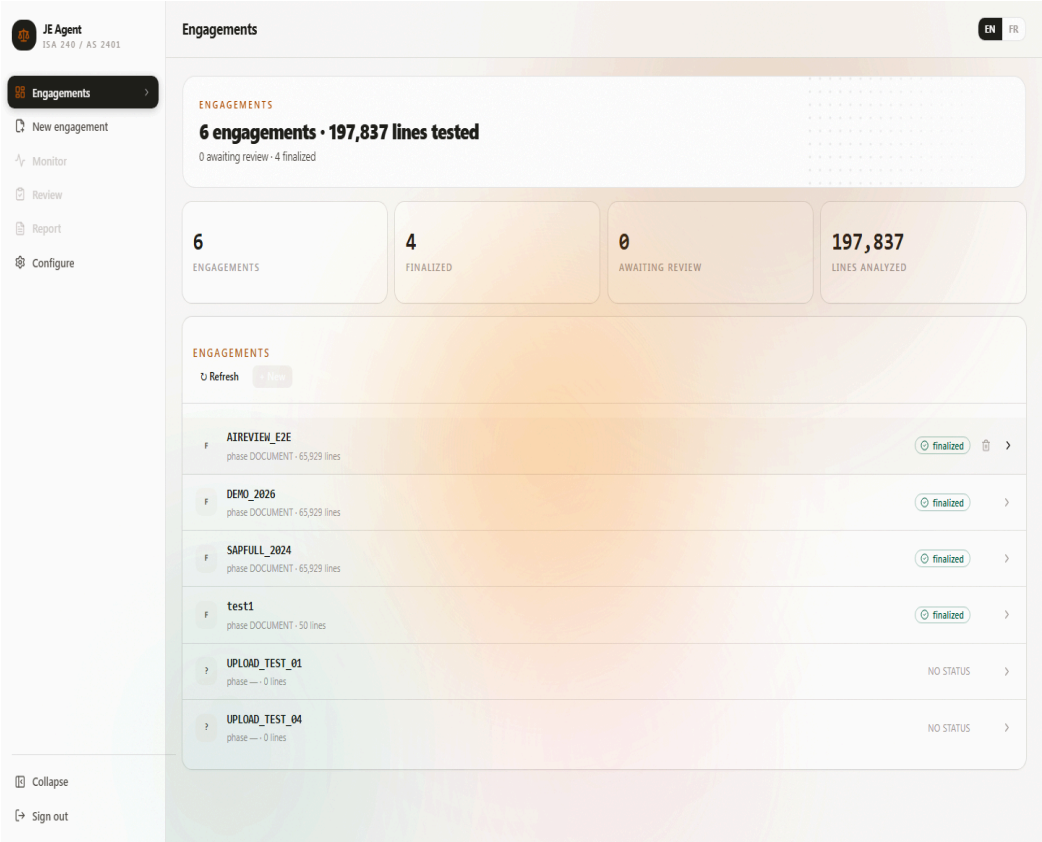
Layer 2, AI triage. A language model now walks through the flagged entries and ranks them: which of these, read together, look most like a story worth a human's evening? It writes a short rationale for each. Its judgment is recorded, labeled as machine advice, and stripped of any authority. The model proposes. It never disposes.

Layer 3, human judgment. Whatever reaches review demands an explicit decision from the auditor: inspect, or accept, with a written reason. That reason is not bureaucracy. It is the workpaper. And every decision is joined to the previous one by a cryptographic fingerprint, forming a chain in which any later tampering would be visible at a glance.

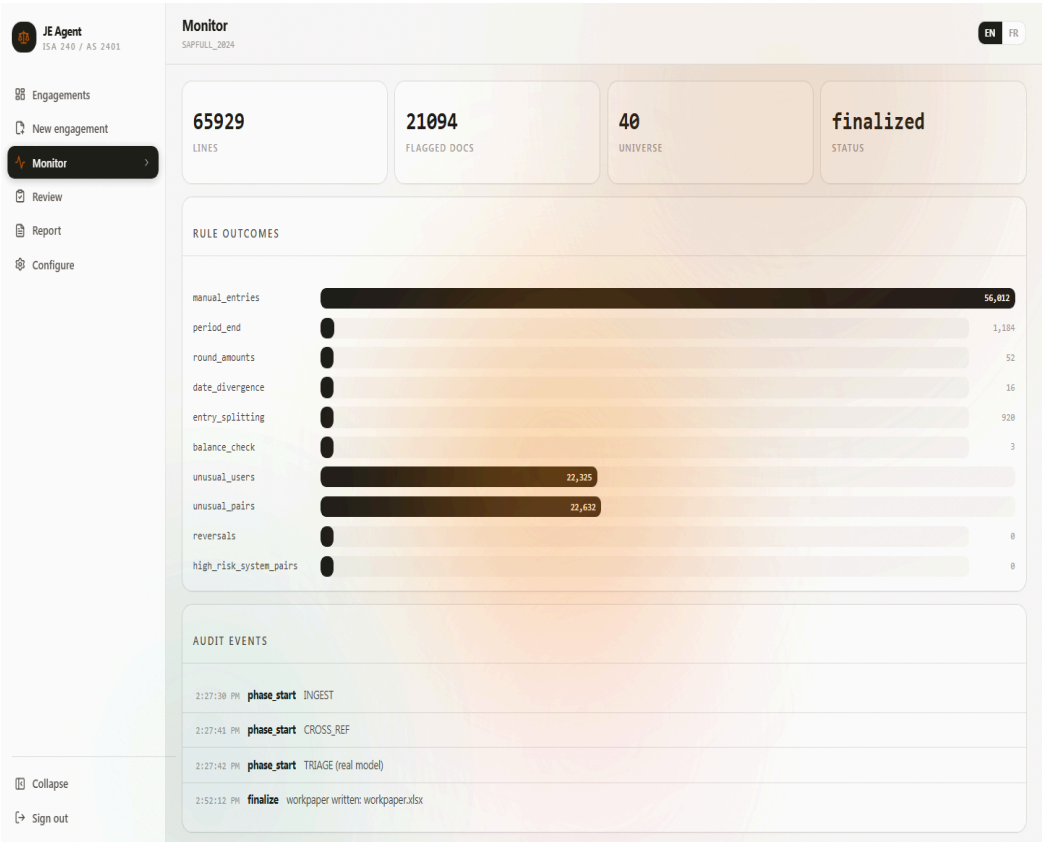
When the review closes, the pipeline produces what an engagement file actually needs: a professional PDF report in English or French, an Excel workpaper, and the list of flagged entries. The machine did the reading. The model did the sorting. The auditor did the deciding. Everyone worked within their strengths.

3 • The Console

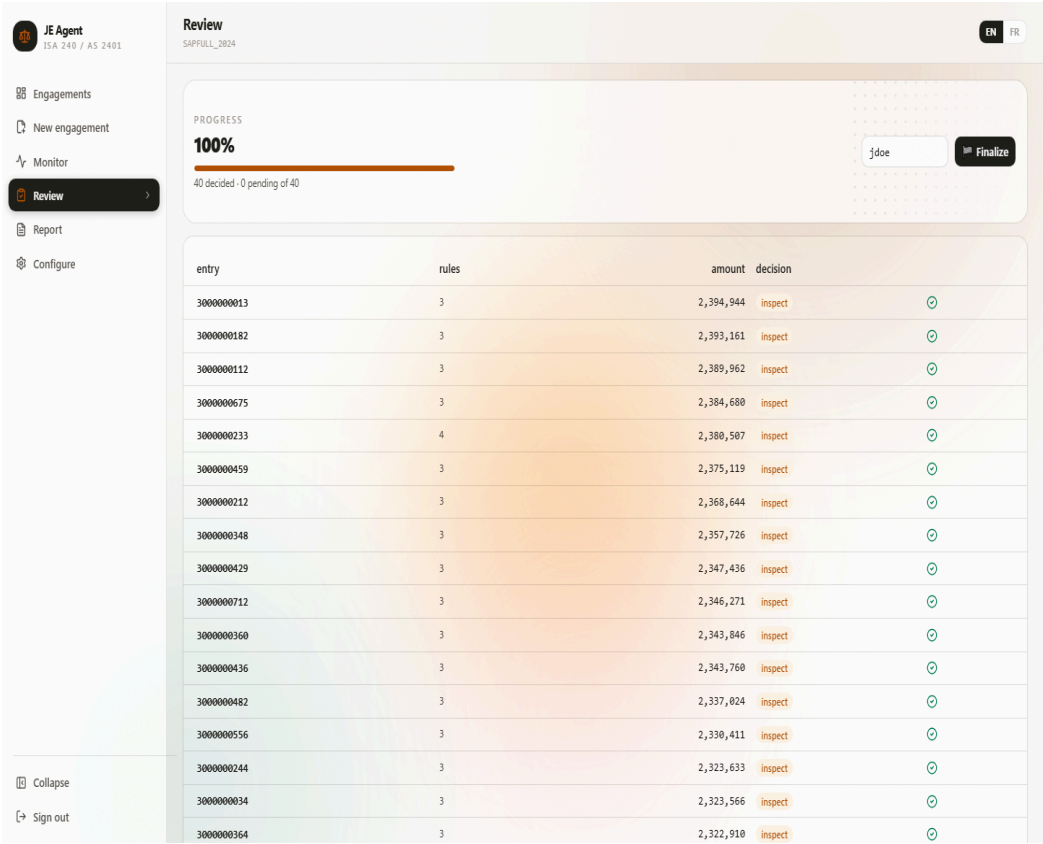
JE Agent runs locally. Client data stays on the machine; only optional triage calls leave it, and those run in zero-retention mode.



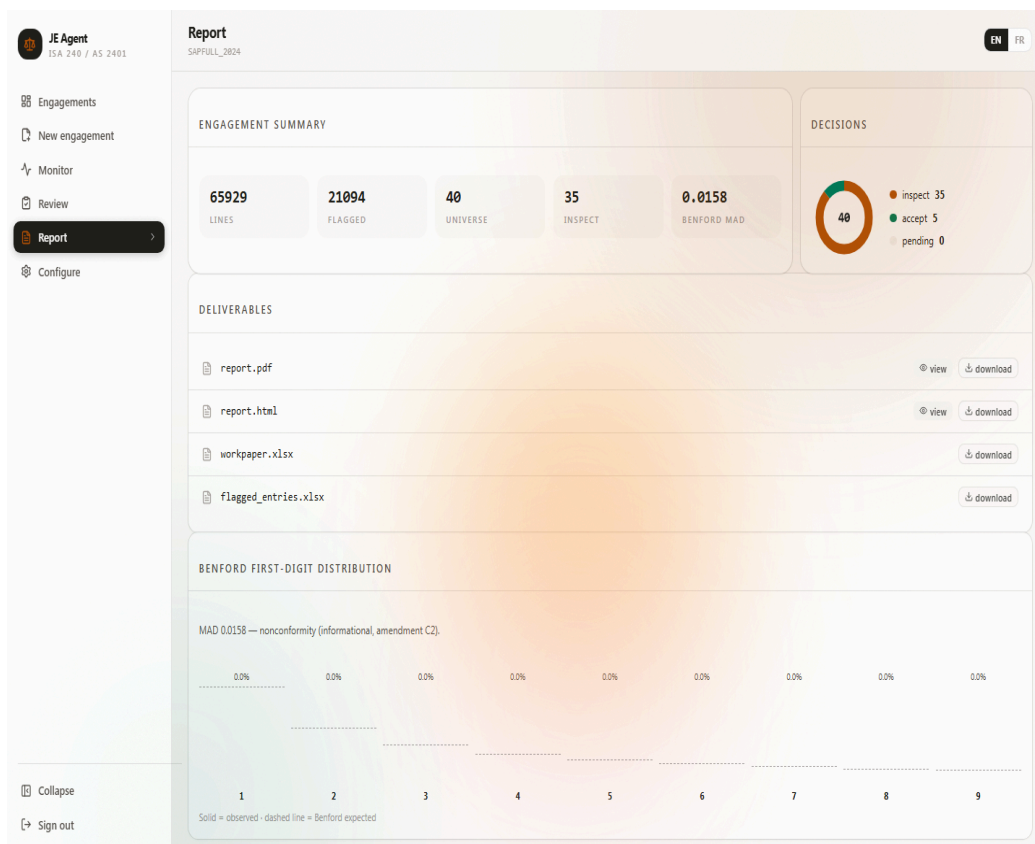
The dashboard is the morning view: every engagement, its population, its status, one line each.



Monitor answers the question an auditor asks mid-run: what did the rules find, and what has the pipeline done so far? Every stage leaves an event in the timeline.



Review is where the engagement is won. Entries appear as a ledger; clicking one pulls out its full journal lines and names the rules that fired. A decision costs one click plus one honest sentence.



The Report page collects the deliverables, breaks down the decisions, and compares first-digit frequencies against Benford’s expectation, offered strictly as information.

JE Agent ISA 240 / AS 2401

New engagement

Drop your journal-entry CSV here
or click to browse - columns auto-detected

CONFIGURATION

ENGAGEMENT

Run ID: DEMO_2026 Period end: 06/30/2026

MATERIALITY

Overall: 250000 Performance: 175000 Currency: USD

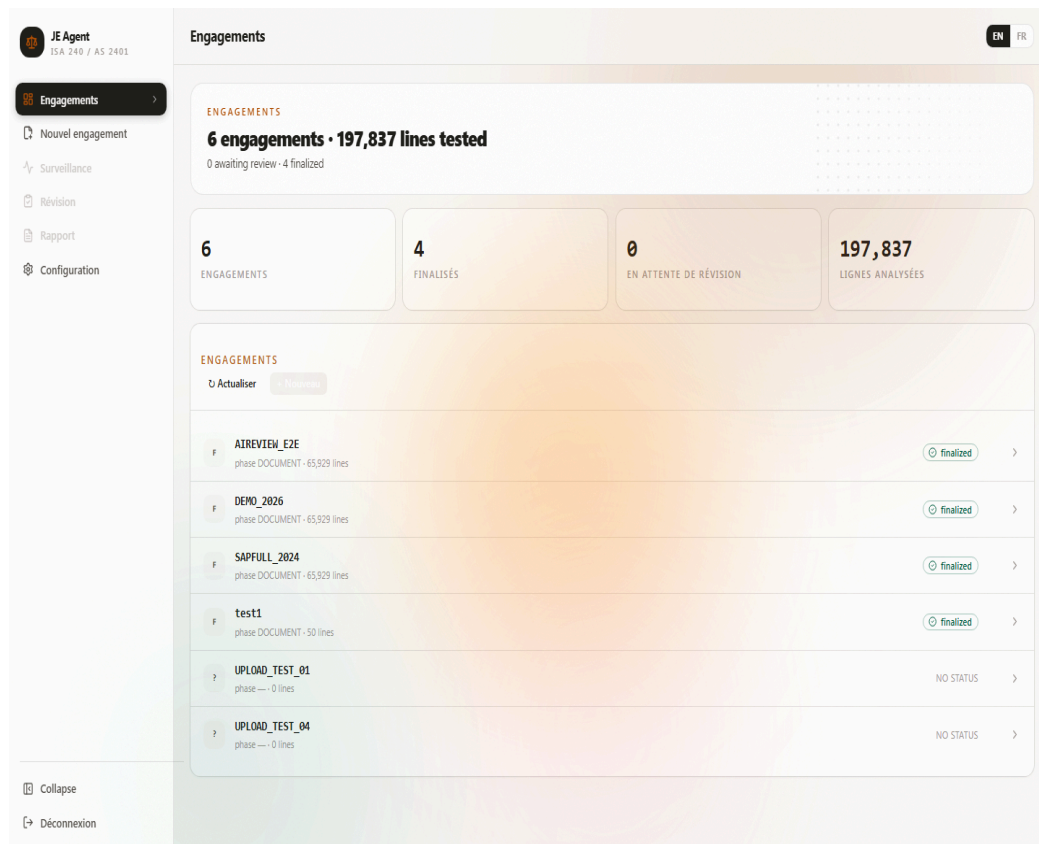
SOURCE SYSTEM

System: SAP Amount column: DMBTR Currency column: WAERS

COLUMN MAPPING

Posting date	Document date	Account
BUDAT	BLDAT	HKONT
Username	Description	Source doc
UNAME	SGTXT	BELNR
Entry ref	Created date	Entry type

Starting work is a CSV dropped into a box. Columns are recognized automatically; materiality and language are chosen in one form.



The interface and the generated report both speak English and French.

4 • Does It Work? The Stress Test

Claims are cheap. This section is about the expensive kind of answer.

To measure detection honestly, you need ground truth, and the real world does not come labeled. So we built the opposite of a demo: a stress harness that manufactures accounting populations with **fraud planted on purpose**. Every planted anomaly carries a label naming the rule that should catch it. Because the truth is known by construction, recall and precision stop being opinions and become arithmetic.

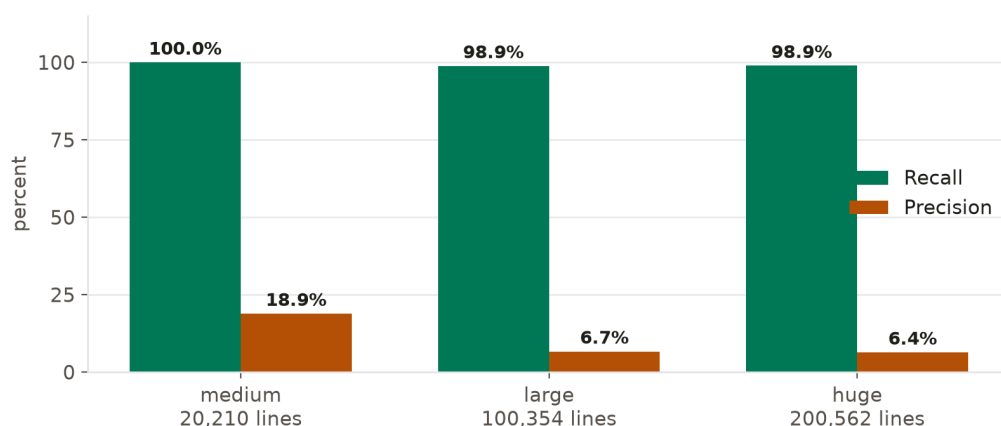
Methodology

Three populations were generated, from twenty thousand to two hundred thousand lines, each deterministic under its random seed. Crucially, the clean base was made deliberately impure: roughly nine tenths manual postings and a tenth system-posted entries, legitimately rare users, legitimately large invoices, month-start postings, the ordinary texture of a real ledger. Noise exists so that false positives *can* happen and therefore can be counted. A test built on sterile data proves nothing except that the test was sterile.

For each rule we score **recall**, the share of planted anomalies caught, and **precision**, the share of flags that were genuinely planted. False positives count only when they land on clean documents.

Headline results

Scenario	Lines	Planted	Recall	Precision
medium	20,210	105	100%	18.9%
large	100,354	177	98.9%	6.7%
huge	200,562	281	98.9%	6.4%

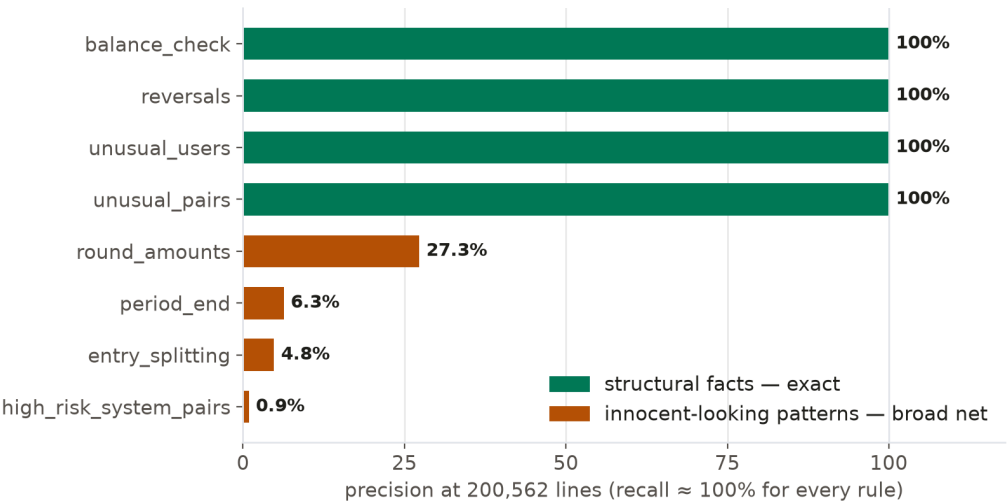


Read these two columns as one story. Recall barely moves: at every scale, virtually nothing planted escapes. Two hundred thousand lines ingest in about fourteen seconds, and the net holds. Precision tells the harder half of the truth. As populations grow, innocent entries that merely resemble fraud grow with them, while

true fraud stays rare by nature. The net catches everything, and everything includes a great deal of harmless fish.

Per-rule picture (200k-line scenario)

Rule	Recall	Precision
round_amounts	100%	27%
entry_splitting	100%	5%
period_end	100%	6%
unusual_pairs	96%	100%
unusual_users	100%	100%
balance_check	100%	100%
reversals	100%	100%
date_divergence	87%	100%
high_risk_system_pairs	100%	0.9%



Two families emerge, and the split teaches more than any single number. Rules grounded in **structural facts**, balance failures, true reversals, users with no history, account pairs that should not exist, are essentially perfect in both directions.

Arithmetic does not have innocent look-alikes. Rules based on **patterns with innocent explanations**, roundness, timing, size clustering, catch every plant while flagging legions of legitimate invoices. They are spotlights pointed at a crowd: everyone suspicious is lit, and so is everyone else.

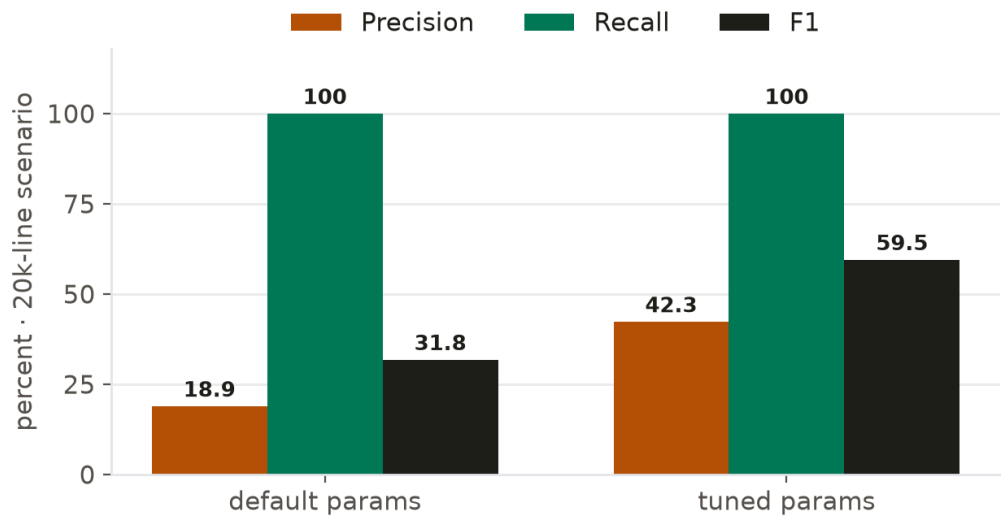
The universe check

No reviewer reads every flag. The system selects a *universe*, ranked and scoped to entries above performance materiality. In testing, between ninety-seven and one hundred percent of above-materiality plants reached the reviewer. The scoping logic holds its end of the bargain.

Can tuning help? Yes, measurably.

Thresholds are not sacred constants, so we treated them as hypotheses. On the twenty-thousand-line scenario, two changes were tested against each other: a narrower period-end window and a stricter rarity definition for system accounts.

Metric	Default	Tuned
Aggregate precision	18.9%	42.3%
Aggregate recall	100%	100%



Precision more than doubled while recall held perfectly. The broader lesson matters more than the numbers: thresholds interact with the population they face, so they must be calibrated against each engagement's own materiality. The harness turns that calibration from folklore into measurement.

The AI layer, scored separately

On the smallest scenario the real triage model ran against ground truth. It caught **one hundred percent of planted anomalies**, and recommended inspecting many clean entries besides, an inspect precision near forty-one percent. Read cynically, that sounds weak. Read correctly, it is the design working: the model is the broad prioritizer feeding a human, not the decider. Each layer missed nothing, ranked honestly, or decided responsibly, exactly one job each.

5 • Trust & Integrity

An audit tool with a malleable trail is a liability wearing a feature's clothing. Four guarantees anchor JE Agent.

Hash-chained decisions. Every reviewer decision fingerprints the one before it. Rewrite history anywhere and the chain snaps visibly.

Citation gates. The generated narrative may not finalize unless each claim cites verifiable data from the run. The report cannot say what the run does not show.

Deterministic core. Same input, same flags, every time. Results can be reproduced years later.

Human-only finalization. Four gates, covering review completeness, documented procedures, valid citations, and stated limitations, block finalization until a named person's decisions are in place. No configuration flag overrides the human.

6 • Honest Limitations

A tool that hides its weaknesses is selling something. These are measured, published weaknesses.

Precision at scale is low by construction, six percent aggregate at two hundred thousand lines. The rule layer is a sieve, and the sieve is the point: fraud is rare and dressed as normality. But users must expect to review, not to receive verdicts.

One rule depends on configuration. High-risk users must be declared for `high_risk_system_pairs` to mean anything; unconfigured, it measured 0.9 percent precision, and we print that number rather than bury it.

`date_divergence` missed about thirteen percent of plants whose forged document dates landed inside the benign window. A known edge, kept on the record.

The populations, however realistically textured, are synthetic. Live client data will surprise the harness the way it surprises everything else.

7 • Conclusion

JE Agent set out to reconcile three things that usually fight each other: speed, transparency, and honesty. The evidence in this report says the reconciliation held. Near-perfect recall at two-hundred-thousand-line scale, delivered in seconds. An architecture in which the AI holds no final authority. An audit trail engineered to expose tampering. And limitations measured in the open, printed beside the strengths rather than beneath a marketing fold.

That last property is the product's real thesis. A tool that tells you exactly what it is good at, exactly where it fails, and shows the arithmetic for both, has earned the

right to sit next to a professional signature.

Appendix A · The ten rules in plain language

#	Rule	What it looks for	Why it matters
1	manual_entries	entries typed by hand rather than posted by a system	manual entries are where override risk lives
2	round_amounts	amounts on suspiciously round numbers (10,000 / 100,000...)	invented figures are rarely precise
3	entry_splitting	one obligation broken into pieces just under approval limits	classic way to dodge a second signature
4	period_end	entries posted in the window around period close	closing pressure is when numbers get “managed”
5	unusual_pairs	account combinations that rarely occur together	e.g. revenue debited against an asset account
6	unusual_users	user IDs with almost no posting history acting alone	ghost or borrowed credentials leave little trace
7	balance_check	documents whose debits and credits don’t reconcile	unbalanced postings indicate broken controls
8	reversals	an entry reversed shortly after posting, same amount	reversal after benefit = possible concealment

#	Rule	What it looks for	Why it matters
9	date_divergence	document date far from posting date	backdating shifts transactions into wrong periods
10	high_risk_system_pairs	configured high-risk users combined with unusual accounts	privileged users need scrutiny wherever they touch

Every rule is a transparent, inspectable check. No black boxes operate at this layer.

Appendix B · Running it yourself

JE Agent runs entirely on your machine. Three commands from a fresh clone:

```

1  git clone https://github.com/salaheddine-11/je-agent && cd je-agent
2  uv sync
3  uv run jeagent serve

```

Then open <http://localhost:8300> in a browser (default key: jeagent).

A typical engagement takes about five minutes of attention:

1. **New engagement:** drop in your CSV export; column mapping is detected automatically. Set overall/performance materiality, choose human or AI-assisted review mode, pick report language.
2. **Monitor:** watch the pipeline stages verify themselves: ingest → rules → cross-reference → triage.
3. **Review:** work the ledger. Every entry shows which rules fired and its full journal lines; each decision needs one click plus a written reason.
4. **Finalize:** four gates check review completeness, procedures, citations and limitations; then the PDF report, Excel workpaper and flagged-entry list are

generated.

Requirements: any modern OS, Python 3.12+, ~200 MB free space. Internet is needed only for the optional AI triage calls (zero-retention mode); everything else runs locally, so client data never leaves the machine.

Glossary

Journal entry (JE): the atomic record of an accounting transaction: dated lines, each crediting or debiting an account.

ISA 240 / AS 2401: international (IAASB) and US (PCAOB) standards requiring auditors to address fraud risk, explicitly including journal-entry testing.

Materiality: the size threshold above which misstatements matter to the financial statements. JE Agent distinguishes *overall* materiality from *performance* materiality (the working threshold for testing).

Performance materiality (PM): see above; entries above PM drive the review universe's scope.

Universe: the ranked selection of flagged entries the reviewer actually sees, scoped to items above performance materiality and capped for practical review.

Recall: of all true anomalies present, the share the system caught. Missing fraud is the costly error, so recall is the headline metric.

Precision: of everything the system flagged, the share that was genuinely anomalous. Low precision means noise the reviewer must wade through.

F1: the balanced average of recall and precision.

Benford's law: natural data has predictable first-digit frequencies (1 is far more common than 9). Fabricated numbers drift away from this pattern; JE Agent reports the comparison as an informational indicator only.

Hash chain: each decision record carries a cryptographic fingerprint of the previous one, so altering history breaks the chain and becomes detectable.

Zero retention: AI calls configured so the provider stores nothing about the request contents.